

Evaluate Hydraulic Fracturing Effectiveness by Combining Water Hammer Analysis and Unified Transient Analysis: A Case Study of Gulong Shale Oil

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Summary

Accurate evaluation of hydraulic-fracturing (HF) effectiveness in shale oil reservoirs remains challenging because of complex induced fracture networks and limited access to low-cost, high-resolution diagnostics. In this study, we present an integrated, data-driven workflow that combines water hammer analysis (WHA) and unified transient analysis (UTA) to quantify both stimulated fracture geometry and the production-contributing fracture surface area from routinely available shut-in pressure and production records. WHA interprets early-time shut-in pressure oscillations to infer fracture complexity and characteristic dimensions, while UTA uses superposition principle and slope analysis to estimate the effective fracture surface area during production. Field case study indicates that the fracture surface area estimated by WHA is less than half of that estimated by UTA, indicating that a large fraction of the stimulated volume may not contribute to sustained flow. Correlating WHA and UTA results with logging data and operational parameters helps identify key drivers of fracture usage and provides actionable guidance to refine stimulation designs in data-limited shale plays.

Introduction

HF is an essential technique for unlocking hydrocarbon resources in unconventional reservoirs, where low permeability restricts natural fluid flow. By injecting large volumes of water, proppants, and chemical additives, HF creates and maintains high-conductivity fractures that enhance hydrocarbon migration from the reservoir to the wellbore (Economides and Martin 2007; Tang et al. 2011). However, as unconventional resource development expands, accurately evaluating HF performance and optimizing treatment designs are becoming critical for maximizing recovery, minimizing environmental impact, and managing operational costs.

The evaluation of HF-induced fractures traditionally relies on mechanical models, numerical simulations, and field-based monitoring techniques. Classical models, including the Perkins-Kern-Nordgren and Khristianovich-Geertsma-de Klerk models, have long been utilized to estimate fracture dimensions and propagation characteristics based on fundamental rock mechanics (Perkins and Kern 1961; Geertsma and De Klerk 1969; Nordgren 1972). With advances in computational methods, these models have evolved into pseudo-3D, planar-3D, and fully 3D approaches, offering improved accuracy in fracture characterization (Adachi et al. 2007; Zhuang et al. 2016). Nevertheless, these models often require extensive computational resources and lack the capability for real-time analysis, making them impractical during live HF operations in the field (Shen et al. 1997).

Beyond model-based approaches, real-time field monitoring techniques, such as microseismic and distributed fiber-optic sensing (i.e., distributed temperature sensing and distributed acoustic sensing), have emerged as key tools for HF evaluation. Microseismic monitoring captures seismic events associated with fracturing, generating 3D fracture maps that reveal fracture orientation, length, and height. However, it suffers from resolution limitations, increasing uncertainty in complex fracture networks, and requires costly downhole monitoring equipment (Li et al. 2021a; Fang et al. 2022). Distributed fiber-optic sensing provides real-time acoustic and thermal data along the wellbore, aiding in the detection of fracture initiation and fluid flow patterns; however, the high installation and data-processing costs limit its widespread application in HF projects (Huang and Liu 2017). These challenges have prompted the search for alternative or complementary methods that are both cost-effective and applicable to real-time operations.

WHA has recently emerged as a promising alternative for real-time fracture diagnostics. The water hammer effect, which generates transient pressure waves following sudden changes in flow within a closed system, offers insights into fracture characteristics when analyzed during HF operations (Holzhausen and Gooch 1985). During HF, abrupt pump shutoff induces pressure pulses that travel through the wellbore and dissipate within the reservoir. This phenomenon allows for the estimation of fracture length, complexity, and potentially proppant placement by analyzing the wave behavior. Mondal (2010) introduced a semi-analytical model using the method of characteristics to quantify the water hammer effect, which incorporates wellbore and reservoir impedance to predict fracture properties. Later, Carey et al. (2015) expanded this model, demonstrating that water hammer signals can reveal critical fracture parameters, including fracture length and complexity, particularly in scenarios involving multiple fractures. While WHA provides valuable insights, it may not capture all fracture dynamics comprehensively.

Traditional analysis methods, such as rate transient analysis (RTA) and pressure transient analysis (PTA), are widely used for well performance evaluation. RTA primarily focuses on analyzing production data to infer formation permeability and fracture effectiveness over time (Wattenbarger et al. 1998; El-Banbi and Wattenbarger 1998). However, RTA requires a sufficiently long production history and is highly sensitive to production rate variations, making it difficult to apply in early production stages. Moreover, RTA alone cannot provide direct information on reservoir pressure, which limits its ability to accurately characterize early-time reservoir behavior. PTA, on the other hand, relies on shut-in pressure buildup tests to estimate reservoir properties. It provides crucial insights into reservoir connectivity and formation permeability but is limited by the need for controlled shut-in conditions. Furthermore, in unconventional reservoirs where fluid mobility is low, pressure buildup tests may take an extended period to yield meaningful results, delaying decision-making processes

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(Anderson et al. 2010; Chu et al. 2017). To overcome these limitations, UTA addresses this limitation by integrating RTA and into a single framework. Unlike conventional methods that treat production and shut-in data separately, UTA utilizes all available transient data in a unified manner, allowing for a more accurate evaluation of reservoir and fracture properties without requiring prior knowledge of initial reservoir pressure (Wang et al. 2022). This approach enables a holistic interpretation of flow regimes and fracture behavior, particularly in unconventional plays where traditional diagnostic methods may be inadequate. The UTA provides a unique advantage in estimating the effective fracture surface area that actively contributes to hydrocarbon production. By integrating production and shut-in data, UTA captures the dynamic behavior of the reservoir and fractures, focusing on the portion of the fracture network that effectively contributes to flow. During production, the pressure drop across the fracture network reflects the connectivity and conductivity of the fractures, while shut-in data provides additional insights into the reservoir's ability to recharge the fractures. This holistic approach ensures that only the fractures contributing to production are accounted for, offering critical insights for optimizing HF designs and improving well performance.

Our aim with this study is to evaluate the effectiveness of HF in the Gulong shale oil field by integrating WHA and UTA, both applied to the same horizontal well. WHA is used to estimate stage-level fracture geometry and oscillation behavior immediately after shut-in, while UTA is used to quantify the dynamic flow contributing area over time. The analysis reveals that fracture length and complexity exhibit moderate positive correlations with gamma ray index, indicating that more organic-rich and clay-rich intervals tend to generate longer and more complex fractures. Furthermore, the effective fracture surface area derived from UTA accounts for only 47% of the total area estimated by WHA, implying that a substantial portion of the stimulated volume does not contribute to production. By jointly interpreting WHA and UTA results, we can assess what proportion of the created fracture area remains conductive under specific treatment and operational conditions, offering valuable guidance for optimizing stimulation design and improving fracture use efficiency.

Gulong Shale Oil Field

Gulong Shale Oil Formation Overview. The Gulong shale oil formation, located in the Songliao Basin in northeastern China, represents one of the most prolific unconventional oil reservoirs in the country, with complex geological characteristics that make it both a significant resource and a technical challenge for HF (Sun et al. 2021, 2023; Ji et al. 2024). The Songliao Basin, covering approximately 260,000 km², is the largest continental hydrocarbon basin in China and has undergone four major tectonic evolution stages since the Jurassic, including thermal uplift and rifting, extensional faulting, thermal subsidence, and tectonic inversion (**Fig. 1a**) (Li et al. 2021b). These tectonic processes, along with varying sedimentation rates, have contributed to the basin's diverse structural units, which include the northern and southern depressions, northeastern and southeastern uplifts, and the central depression—the primary location of the Gulong shale oil deposits (**Fig. 1b**) (Wang et al. 2016).

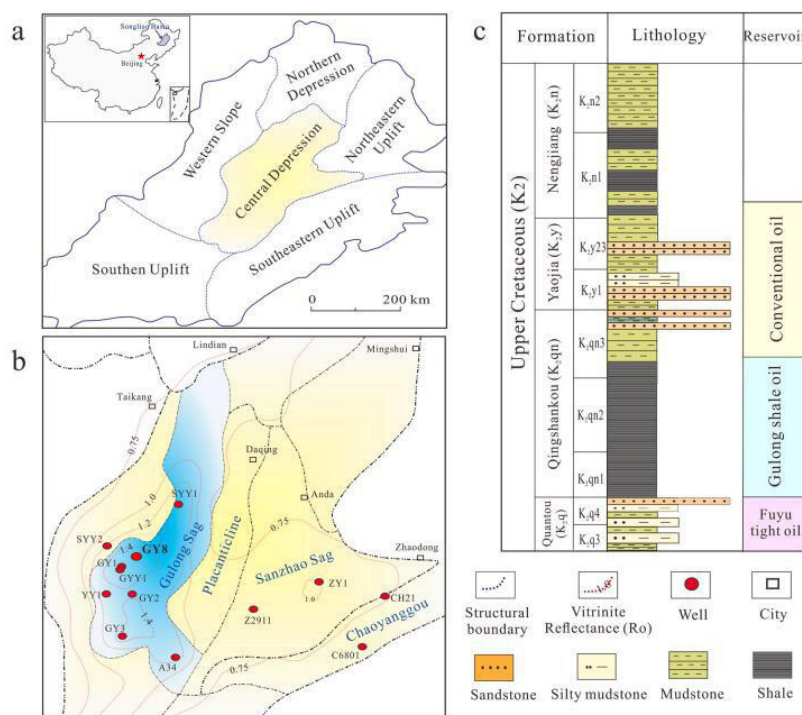


Fig. 1—(a) Location of the central depression, (b) shale oil wells distributed in the Gulong Sag, and (c) stratigraphic column of the Upper Cretaceous Formation in the Songliao Basin. The contours of vitrinite reflectance (Ro) of the K2qn1 organic matter (OM) were modified after Sun et al. (2021).

Composed mainly of the Cretaceous Qingshankou Formation, the Gulong shale oil reservoir is characterized by high organic content, thick shale intervals, and complex pore structures. With shale content exceeding 95%, the formation exhibits interbedded layers of dolomite, shell limestone, and siltstone, each only a few centimeters thick, contributing to its low permeability and significant heterogeneity (**Fig. 1c**) (Feng et al. 2010). The clay mineral content, which ranges from 25 to 40% with an average of 35.6%, primarily consists of illite with minor amounts of montmorillonite below 1650 m depth (Zhang 2021). The presence of these clay minerals influences fracture propagation, as higher clay content often correlates with increased ductility, potentially limiting fracture extension under hydraulic stimulation.

Pore structures in the formation are intricate, ranging from nano- to micron-scale pores combined with microfractures. Matrix pores, typically ranging from 10 nm to 50 nm in diameter, include intergranular mineral pores, clay pores, organic matter pores, and dissolution pores, with a median pore size between 20 nm and 30 nm (Sun et al. 2024). Field emission scanning electron microscopy has revealed that bedding fractures are approximately 50–150 nm wide, while intergranular pores and microfractures form a network that facilitates fluid flow within the shale matrix. This intricate pore system, combined with the high organic matter content, presents a favorable environment for hydrocarbon accumulation but also poses challenges for efficient fracture connectivity and fluid extraction.

Stratigraphic sequences in the Songliao Basin range from 3000 m to 7000 m thick, comprising formations from the Triassic through the Neogene, overlaying a metamorphic basement. The Gulong shale is particularly notable for its high total organic carbon content, averaging 2.35%, which provides a significant hydrocarbon source. Additionally, the brittle mineral components, including dolomite and calcite, contribute to the brittleness index of approximately 50%, making the shale moderately responsive to fracturing. The high brittleness index aids in fracture initiation, while the clay-rich, ductile interbeds may restrict fracture propagation, resulting in a complex fracture network essential for enhancing well connectivity and production efficiency (Liu et al. 2019; Sun et al. 2023).

The combination of these unique geological properties within the Gulong shale oil formation necessitates tailored HF approaches to optimize hydrocarbon recovery. Understanding the interplay between clay and brittle minerals, pore structure, and organic matter distribution is crucial for designing effective fracturing treatments that maximize fracture connectivity and stimulate productive fractures. In this study, insights from WHA and UTA provide critical evaluations of fracture characteristics in this complex reservoir, guiding strategies for optimizing production from the Gulong shale oil formation.

A-Well Overview. A-Well is a representative horizontal producer within the Gulong Sag of the central depression. The well targets the Q9 member of the Qingshankou Formation and was drilled to a measured depth of 4630 m with a long horizontal section in Q9. Petrophysical partitioning defines 21 pay subintervals totaling 2080 m, including 16 Type I-1 layers (2000 m cumulative length) and five Type I-2 layers (78.9 m), reflecting a high-quality but laminated reservoir architecture (Sun et al. 2023). Core- and log-constrained rock physics indicate an average brittleness index near 51% with effective porosity of 7–8%, consistent with moderate-to-high fracturability while preserving sufficient matrix storage (Li et al. 2021b).

Reservoir and fluid conditions used in subsequent modeling are derived from high-pressure pressure/volume/temperature in adjacent Q9 wells and on-site measurements: in-situ oil density of 0.564 g/cm³, oil viscosity of 0.26 mPa·s, formation volume factor of 1.9, gas/oil ratio of 397.36 m³/t, and bubblepoint pressure of 30.811 MPa. The pressure coefficient is 1.51, indicating an overpressured system favorable for sustained conductivity after stimulation (Liu et al. 2019).

Completion used casedhole multistage HF with 45 stages and 178 clusters to maximize reservoir contact within Q9 (Zhang 2021). Shut-in and early flowback were monitored with an automatic surface pressure logger synchronized with production rate records, yielding good quality data required by both methods: WHA interprets the immediate shut-in oscillatory response to infer fracture complexity and a stimulated surface area $A_{f,WHA}$; UTA interprets the variable rate production history to estimate the effective contributing fracture area $A_{f,UTA}$ from linearflow slope analysis.

A pressure decline and water hammer data example is provided in **Fig. 2**. These figures illustrate the multistage fracturing layout, pressure response, and production trends observed in Stage 1, serving as representative examples for multistage fracturing response across the well. **Fig. 3** further presents production data, showing both the time vs. production plot and the time vs. pressure plot. These figures collectively illustrate the well's production trends and the relationship between pressure and production rates across different stages, providing essential data for subsequent analysis. The pressure and production data will be further analyzed using WHA and UTA to evaluate fracture effectiveness and refine production optimization strategies for the Gulong shale oil field.

Methodology

This section briefly introduces the fundamentals of WHA and UTA, which form the basis for subsequent fracture characterization.

WHA. The transient behavior of pressure in the wellbore is governed by 1D equations of motion and continuity (Wylie et al. 1993).

$$\frac{\partial Q}{\partial t} + gA \frac{\partial H}{\partial x} + \frac{fQ|Q|}{2DA} = 0, \quad (1)$$

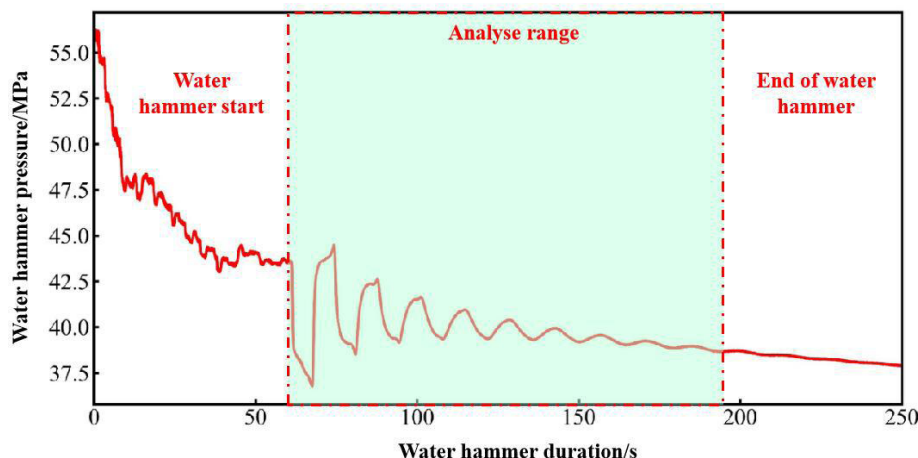


Fig. 2—Pressure decline and water hammer data for Stage 1 of A-Well.

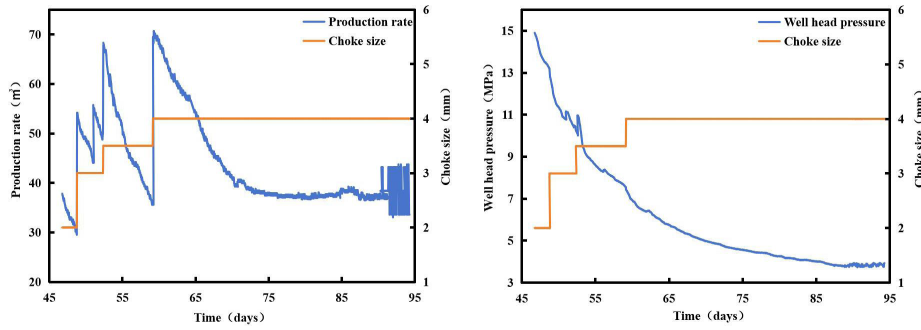


Fig. 3—(Left) Time vs. production; (right) time vs. pressure.

$$\frac{\partial H}{\partial t} + \frac{a^2}{gA} \frac{\partial Q}{\partial x} = 0, \quad (2)$$

$$H = \frac{P}{\rho g} + z, \quad (3)$$

where a is the wave velocity, f is the Darcy-Weisbach friction factor, A is the pipe cross-sectional area, Q is the flow rate, D is the pipe diameter, and H is the pressure head.

The water hammer wave velocity in the wellbore is expressed by Halliwell (1963) as

$$a = \sqrt{\frac{K/\rho}{1 + (K/E)\psi}}, \quad (4)$$

where K is the bulk modulus, ρ is the density, E is the Young's modulus of elasticity of the pipe, and ψ depends on a dimensionless variable of the elastic properties of the pipe.

$$\psi = \frac{D}{e} (1 - \nu^2), \quad (5)$$

where ν is the pipe Poisson's ratio and e is the wall thickness.

Eqs. 1 and 2 of the water hammer equation are a pair of proposed linear hyperbolic partial differential equations, which can only be solved numerically. Therefore, the system of partial differential equations is transformed into four ordinary differential equations using the method of characteristic lines method, and then the system of equations is solved using the finite difference method, and the characteristic lines C^+ and C^- equations are represented by Eqs. 5 and 6 (Wylie et al. 1993).

$$C^+: H_i^{t+\Delta t} = H_{i-1}^t + B \left(Q_i^{t+\Delta t} - Q_{i-1}^t \right) - r Q_{i-1}^t |Q_{i-1}^t|, \quad (6)$$

$$C^-: H_i^{t+\Delta t} = H_{i+1}^t + B \left(Q_i^{t+\Delta t} - Q_{i+1}^t \right) - r Q_{i+1}^t |Q_{i+1}^t|, \quad (7)$$

where B is the pipe characteristic impedance, given by $B = \frac{a}{gA}$; r is the pipe resistance coefficient, given by $r = \frac{f\Delta x}{2gDA^3}$; the wellbore system was divided into n equal reaches or subsections, each Δx in length.

Eqs. 5 and 6 are valid only when both C^+ and C^- equations are present, whereas there is only one equation, C^+ or C^- , for the wellhead and bottomhole, so boundary conditions need to be added to solve for H and Q at the wellhead and bottoming (Wylie et al. 1993).

Boundary Conditions at the Wellhead. The boundary conditions for the fracture-containing water hammer model are divided into two main parts, which are the wellhead boundary conditions and the fracture boundary conditions at the bottom of the well. At the wellhead, the flow rate at the wellhead is assumed to decrease linearly to zero during pump shutdown. Therefore, the boundary conditions at the wellhead can be defined in terms of the change in flow rate. Based on the wellhead boundary flow, the wellhead boundary head can be calculated by the C^- equation.

Wellbore Fracture Boundary Condition. Fracture boundary conditions are typically modeled as either single-fracture or multifracture systems.

In the single-fracture case, according to the description of the wellbore fracture boundary by Carey et al. (2015), in the wellbore fracture boundary condition, the wellbore fracture boundary condition can consist of a fixed resistance R , capacitance C , and inductance I of an equivalent circuit (RCI model).

The boundary condition is denoted as

$$\rho g \Delta H = RQ + \frac{1}{C} \int Q dt + I \frac{dQ}{dt}. \quad (8)$$

The relationship between R , C , and I is shown in Eqs. 9 through 12 (Schönfeld 1954; Holzhausen and Gooch 1985; Mondal 2010; Feynman et al. 2015).

$$R = \frac{\Delta P_{mwf}}{Q_0}, \quad (9)$$

$$C = \frac{\Delta V}{\Delta P}, \quad (10)$$

$$Q = C \frac{dP}{dt}, \quad (11)$$

$$\Delta P = RQ = I \frac{dQ}{dt}, \quad (12)$$

where Q_0 is the injection rate before pump shutdown, ΔP_{nwf} is the near-well friction pressure drop; ΔV is the fracture volume change, and ΔP is the fracture pressure change.

The average net pressure ΔP_0 is the difference between the average pressure in the fracture ($P_{BH} - P_{nwf}$) and the minimum horizontal ground stress (S_{Hmin}).

$$\Delta P_0 = (P_{BH} - \Delta P_{nwf}) - S_{Hmin}. \quad (13)$$

Single-Fracture Model. Shlyapobersky et al. (1988) united the average net pressure with the crack size by the following equation and classified the cracks into long ($2L_f/h_f \geq 1$) and short ($2L_f/h_f < 1$) cracks.

$$\Delta P_0 = \frac{4}{\pi^2} \frac{E}{1-\nu^2} E(m) w \max\left(\frac{1}{L_f}, \frac{2}{L_f}\right). \quad (14)$$

$E(m)$ is an elliptic integral of the second type, and the defining equation for m is shown in Eq. 15.

$$E(m) = \int_0^1 \frac{\sqrt{1-m^2t^2}}{1-t^2} dt, \quad (15)$$

$$m = 1 - \left(\frac{\min\left(L_f, \frac{h_f}{2}\right)}{\max\left(L_f, \frac{h_f}{2}\right)} \right)^2. \quad (16)$$

The formulae for obtaining the length, width, and height of the cracks are shown below (Carey et al. 2015):

$$L_f = \sqrt{\frac{CI\Delta P_0}{\rho}}, \quad (17)$$

$$h_f = \begin{cases} \frac{4E'E(m)C}{\pi^2 L_f^2} & (\text{short fracture}) \\ \sqrt{\frac{4E'E(m)C}{\pi^2 L_f^2}} & (\text{long fracture}) \end{cases}, \quad (18)$$

$$\bar{w} = \frac{\rho L_f}{I h_f}, \quad (19)$$

$$E' = \frac{E}{1-\nu^2}, \quad (20)$$

where ν is the Poisson's ratio.

Multifracture Model. The multifractures boundary condition is based on the single-fracture boundary condition, which accounts for the stress interference effect between multiple fractures (the number of hydraulic fractures equals the number of perforation clusters), and a method to consider the effect of stress interference on fracture modeling is proposed on the basis of the conservation of fracture volume. That is, the ratio of each maximum fracture width is incorporated into the equivalence relation for each fracture. The value of each fracture geometry can then be calculated, and the multiple fracture equivalent circuit model is shown in **Fig. 4**. The R_p , C_p , and I_T of the equivalent single fracture are obtained by fitting the field data (Hu et al. 2022).

Through the RCI circuit, the capacitances of each fracture are connected in parallel, so the capacitance of N fractures can be expressed as

$$C_T = \sum_{i=1}^N C_i, \quad (21)$$

From Eq. 22, the half-length of the equivalent single fracture can be written as the sum of each fracture, and the conversion formula for the fracture half-length can be expressed as

$$L_{fE} = \sqrt{\sum_{i=1}^N L_i^2}. \quad (22)$$

where L_{fE} is the half-length of the equivalent total fracture. Similarly, the equivalent single-fracture width can be written as the sum of each fracture's width. The conversion formula for fracture width can be expressed as

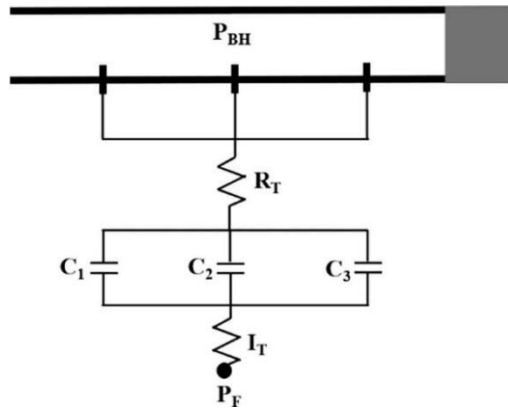


Fig. 4—Schematic of multicrack equivalent circuit (Carey et al. 2015).

$$\bar{w}_E = \sqrt{\sum_i^N \bar{w}_i^2}. \quad (23)$$

According to the assumption that all initial fractures have the same height, the height of each fracture can be defined as

$$h_{fE} = h_{fi} \quad (i = 1, 2, 3 \dots N). \quad (24)$$

Subsequently, the stress interference between each fracture is calculated according to the following equation:

$$\Delta\sigma_{\xi}^{ij} = (P_{BH} - S_{Hmin}) \left[1 - \left(1 + \left(\frac{h_f}{2d_{ij}} \right)^2 \right)^{-\frac{3}{2}} \right]. \quad (25)$$

Among them, $\Delta\sigma_{ij}$ is the stress interference between the i th and j th fractures, and d_{ij} is the fracture spacing between the i th and j th fractures. When calculating the mutual stress interference between fractures, the net pressure of each fracture can be obtained based on the stress interference between fractures. The formula can be expressed as

$$P_{net,i} = P_{BH} - S_{Hmin} \sum_{j=1, j \neq i}^N \Delta\sigma_{\xi}^{ji}, \quad (26)$$

where $P_{net,i}$ is the average net pressure of the i th fracture. It should be noted that if the net pressure of a certain fracture is calculated as a negative value during the calculation process, it is considered that the fracture cannot propagate normally and should be removed, and the net pressure of the remaining fractures should be recalculated. After obtaining the net pressure of all fractures, the fracture width of each fracture under the stress interference model is calculated according to the fracture width expression derived by Sneddon and Elliot, and the formula can be expressed as (Sneddon and Elliot 1946)

$$w_i = \frac{2h_f(1 - \nu^2) P_{net,i}}{E}. \quad (27)$$

Solve for the half-length of each crack according to Eq. 28

$$L_{fi} = \frac{I_T h_{fi}}{\rho w_i}. \quad (28)$$

Among them, I_T and h_{fi} can be derived from the equivalent single fracture. Eq. 14 can be modified to the net pressure of each multifracture.

$$\Delta P_{0i} = \frac{4}{\pi^2} \frac{E}{1 - \nu^2} E(m) \bar{w}_i \left(\frac{1}{L_{fi}}, \frac{2}{h_{fi}} \right). \quad (29)$$

Among them, ν is the Poisson's ratio. The calculated half-length (L_{fi}) and width (w_i) of each of the multiple fractures are used in Eq. 30 to calculate the net pressure (ΔP_{0i}) of each fracture. Finally, Eq. 17 can be modified to the half-length of multiple fractures.

$$L_f = \sqrt{\frac{C_i I_T \Delta P_{0i}}{\rho}}. \quad (30)$$

Then, Eq. 30 can be transformed into

$$C_i = \frac{h_{fi}^2 \rho}{I_T \Delta P_{0i}}. \quad (31)$$

C_i of each fracture with multiple fractures as well as the half-length L_{fi} , fracture width w_i and fracture height h_{fi} of each fracture can all be calculated.

Carey et al. (2015) suggested that the wellbore fracture boundary condition proposed by Mondal (2010) in the case of single-stage multicluster fracturing is to equate multiple fracture effects into a single fracture. Carey et al. (2015) found a strong correlation between the water hammer model and the fracture lengths explained by microseismicity. Iriarte et al. (2017) investigated the decay rate of water hammer. The energy absorbed by the pressure pulse as it passes through the fracture network was considered, and the larger the fracture network, the greater the water hammer attenuation. Thereafter, Hwang et al. (2017) investigated the effect of each parameter in the RCI model on the water hammer signal and applied the RCI model to multicluster fracturing. Even though WHA is used and compared with other diagnostic methods (Haustveit et al. 2017) during and after hydraulic-fracture propagation, no work has been done that can relate WHA to post-production behavior.

Although the geometry of hydraulic fractures can be complex at the near-wellbore scale, such as in an echelon or branching patterns, multiple studies have demonstrated that on a reservoir scale, fracture geometry is often planar in many cases. Shahri et al. (2021) used high-resolution fiber-optic measurements combined with fracture modeling to demonstrate that fracture swarms in HF treatments often exhibit planar, vertically contained, and parallel geometries across multiple stages. Similar conclusions were drawn from the Hydraulic Fracturing Test Site core-through project, which revealed that induced fractures were predominantly planar and subparallel, reflecting a consistent propagation direction (Gale et al. 2014). Shahri et al. (2021) compared waterfall plot simulations and fiber-optic data, revealing that hydraulic fractures can be very well modeled as nearly linear cracks. Furthermore, microseismic monitoring during HF operations has shown that fractures generally align with the maximum horizontal stress and exhibit planar propagation patterns (Alalli and Zoback 2018). Cao et al. (2025) integrated microseismic data and numerical modeling and found that almost all hydraulic fractures propagate along maximum horizontal stress with an average azimuth of NE 83° in a Gulong shale oil well. Liu et al. (2023) analyzed and tested core samples from the Gulong shale formation, which is characterized by high clay mineral content (average 46.6 %) and strong plasticity, thus favorable to planar fracture geometry. Therefore, while acknowledging small-scale complexity, the assumption of planar fracture geometry remains reasonable for modeling and interpreting fracture behavior in the Gulong shale formation. In A-Well, the number of clusters per stage varies from 3 to 5, except for Stage 1, which contains only a single cluster. Consequently, the multifracture RCI model was applied to all stages except Stage 1, where the single-fracture model was used. This differentiation ensures accurate fracture geometry estimation based on cluster configuration.

UTA. UTA integrates production data and shut-in pressure data to evaluate well performance and fracture effectiveness under transient flow conditions. Under linear flow, the pressure response to variable rate production can be interpreted using the following expression:

$$P_{bh} = P_0 - 25.4 \times 10^3 \frac{B}{A_f} \sqrt{\frac{\mu}{k\phi c_t}} \sum_{i=1}^n (q_i - q_{i-1}) \sqrt{\sum t_i - \sum t_{i-1}}, \quad (32)$$

where A_f is the effective fracture surface area, μ is the fluid viscosity, k is the permeability, ϕ is the porosity, C_t is the total compressibility, and B is the formation volume factor.

To normalize the impact of variable production rates over time, a superposition variable τ , also known as convolution time, is introduced. It is calculated as

$$\tau_i = \sum_{i=1}^n (q_i - q_{i-1}) \sqrt{\sum t_i - \sum t_{i-1}}. \quad (33)$$

This transformation enables the plotting of pressure under variable rate conditions as a function of equivalent constant rate time. When bottomhole pressure is plotted against τ , a straightline segment often emerges under linear flow conditions.

The slope m of this linear segment is related to A_f by

$$m = 25.4 \times 10^3 \frac{B}{A_f} \sqrt{\frac{\mu}{k\phi c_t}}. \quad (34)$$

By identifying the linear portion of the pressure– τ plot and fitting the slope $m = \frac{dP_{bh}}{d\tau}$, the effective fracture surface area A_f can be backcalculated assuming known reservoir parameters. This process forms the core of UTA's capability to quantitatively estimate fracture effectiveness using standard field data. The method is illustrated in **Fig. 5** and offers a cost-effective alternative to image-based or microseismic-based fracture diagnostics. UTA is advantageous over traditional RTA, which only uses production data, as it allows for simultaneous analysis of production and shut-in data, providing a more complete picture of the well's performance (Wang et al. 2021, 2022).

All relevant simulation parameters are summarized in **Table 1**. These include wellbore dimensions, formation properties, fluid characteristics, and RCI circuit elements used in the water hammer and transient analysis. The values listed for resistance (R), capacitance (C), and inductance (I) represent the initial guesses required to start the RCI-based WHA simulation. These values are not measured directly but are assumed based on engineering judgment and used as starting points for iterative calibration. The final fracture dimension estimations are obtained after tuning these parameters to match observed pressure responses.

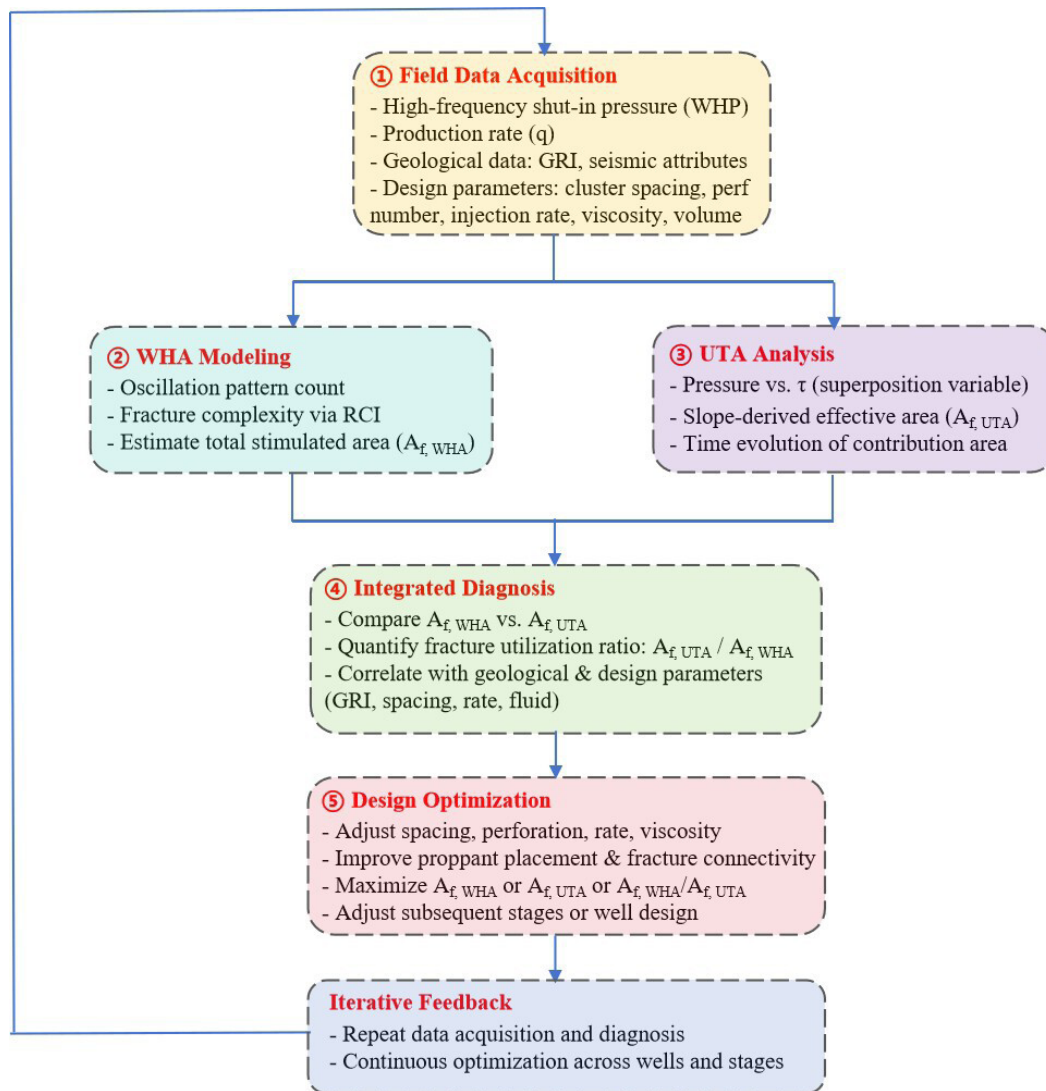


Fig. 5—Workflow for stimulation evaluation and optimization based on integrated WHA and UTA interpretation. WHP = wellhead pressure; GRI =gamma ray index.

Parameter	Value	Parameter	Value
Pipe thickness (m)	0.01	Fluid viscosity (mPa.s)	10
Pipe diameter (m)	0.11	Fluid density (kg/m ³)	1000
Pipe vertical length (m)	4000	Compressibility (1/MPa)	2.09×10 ⁻⁵
Pipe horizontal length (m)	2000	Formation porosity	0.075
Poisson's ratio	0.28	Formation permeability (m ²)	9.87×10 ⁻¹⁹
Young's modulus (GPa)	16.4	Formation volume factor	1.9037
Friction factor	0.05	Minimum horizontal stress (MPa)	51.6
Fracture pressure (MPa)	56.7	Number of clusters per stage	1

Table 1—Input parameters used in WHA and UTA simulations.

To illustrate the calibration process, **Table 2** presents a typical set of initial and final values used for R , C , and I for Stage 1 of A-Well. These values are tuned iteratively during simulation to improve agreement between modeled and observed pressure oscillation responses.

Parameter	Initial Value	Final Value
Resistance (Pa·s/m ³)	10 ⁷	1.9×10 ⁷
Capacitance (m ³ /Pa)	10 ⁻⁵	1.5×10 ⁻⁵
Inductance (Pa·s ² /m ⁶)	10 ⁵	5.8×10 ⁵

Table 2—Initial and final RCI values used in WHA simulation calibration for Stage 1 of A-Well.

The overall diagnostic procedure integrating WHA and UTA is illustrated in Fig. 6. This workflow highlights the key steps in pressure data processing, fracture modeling, and effective area estimation, forming the basis for all subsequent analyses.

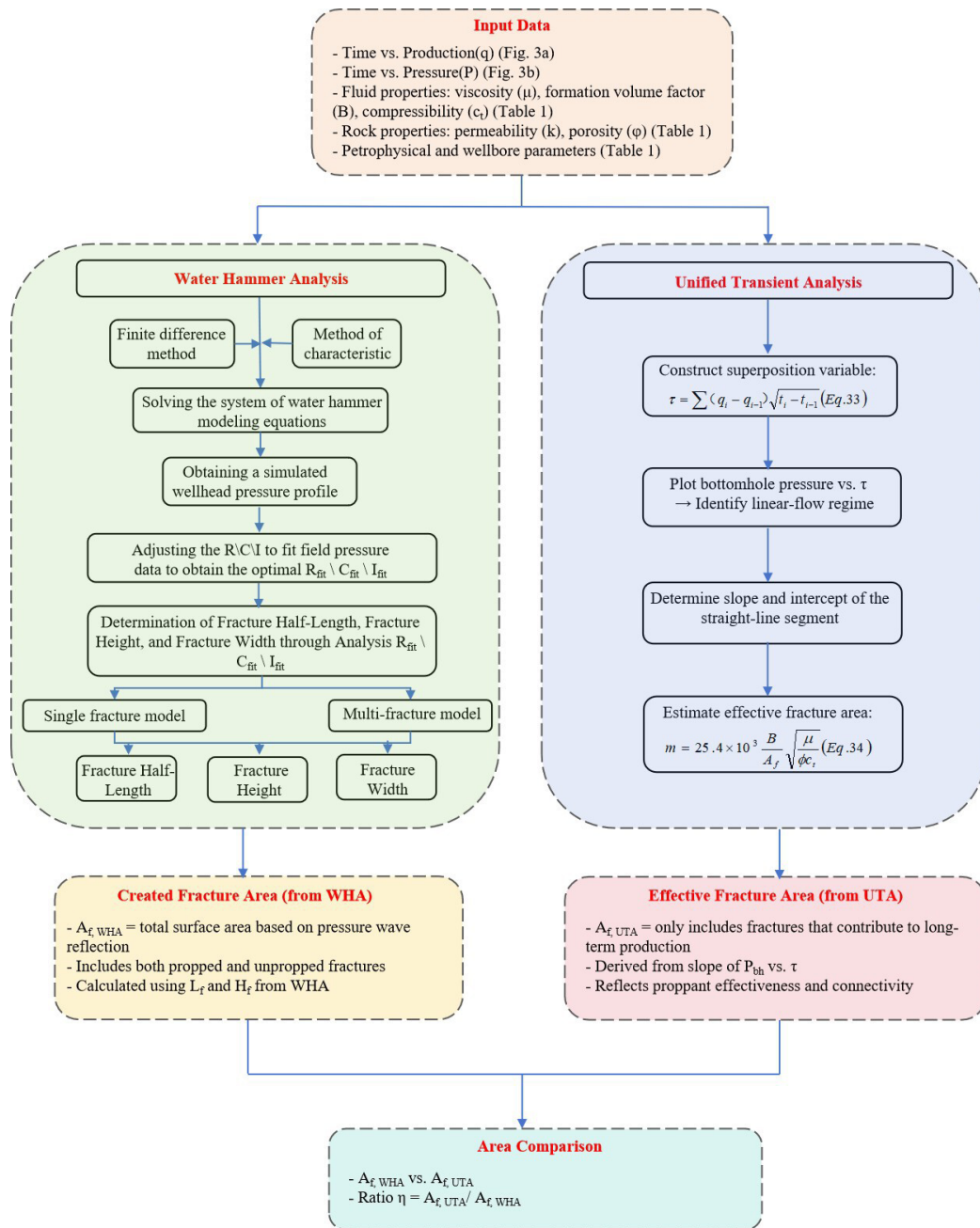


Fig. 6—Workflow for evaluating HF using WHA and UTA.

Analysis and Results

This section presents the results obtained from WHA and UTA, applied to A-Well in the Gulong shale oil field. The focus is on fracture characterization, production effectiveness, and potential optimization strategies.

WHA Results. Water Hammer Model Validation. To verify the reliability of the WHA model, simulated wellhead pressure was compared with field-measured pressure data during the shut-in period. As shown in **Fig. 7**, the simulated pressure curve (red) closely replicates the measured oscillations (black), matching key characteristics, such as peak amplitude, oscillation frequency, and attenuation behavior. This strong agreement validates the WHA model simulation and supports its use for fracture geometry estimation in subsequent stages of A-Well.

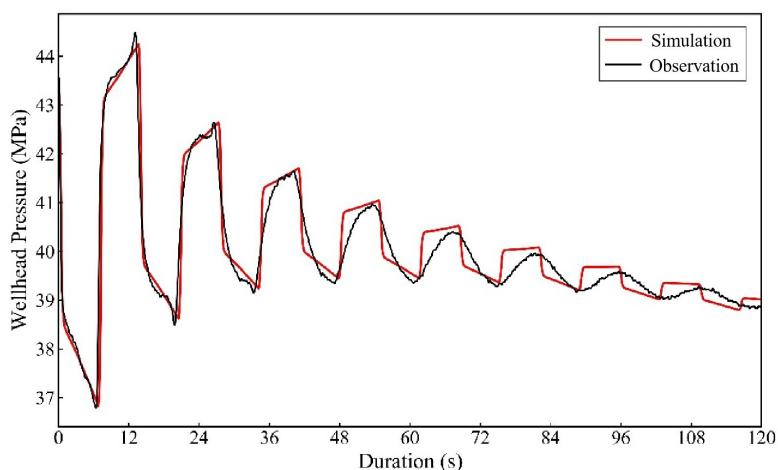


Fig. 7—Comparison between simulated (red) and observed (black) wellhead pressures during HF shut-in for Stage 1 of A-Well.

Fracture Geometry Estimation From WHA. Fracture dimensions for all 45 stages in A-Well were estimated using WHA, under both single-fracture and multifracture model assumptions. The key parameters—fracture half-length and fracture height—are summarized in **Figs. 8 through 10**.

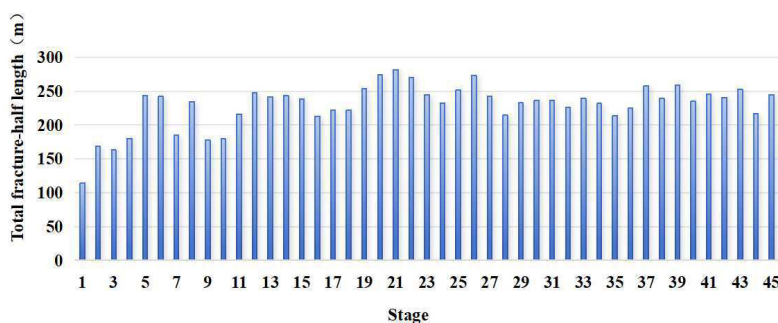


Fig. 8—Total fracture half-length vs. stage diagram.

Total Fracture Half-Length (Single-Fracture Model). As shown in **Fig. 8**, the total fracture half-length, estimated under the single-fracture RCI model assumption, ranged from 114 m (Stage 1) to 281 m (Stage 21), with an average value of 228 m. The significant

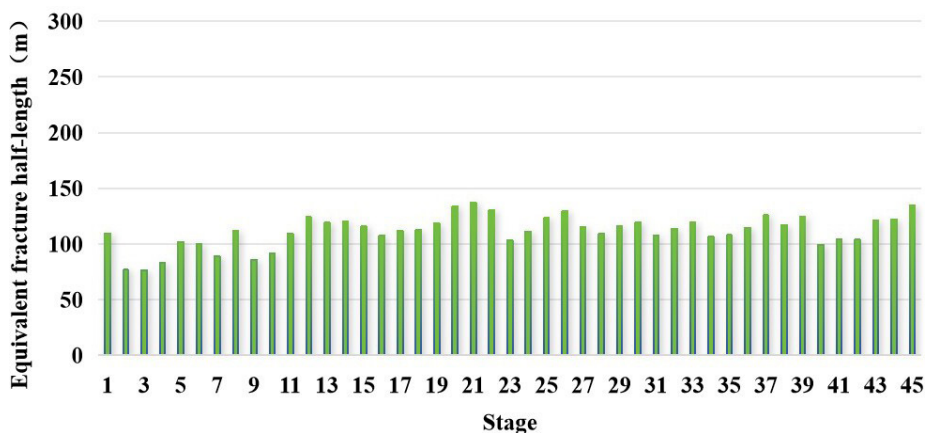


Fig. 9—Per-cluster equivalent fracture half-length vs. stage diagram.

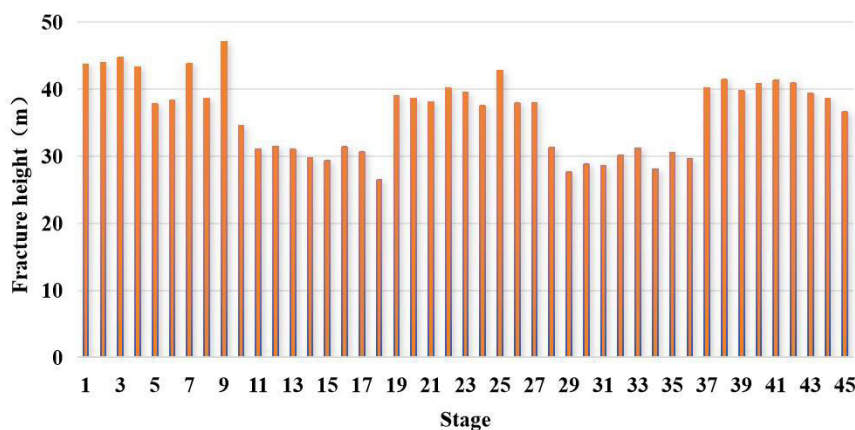


Fig. 10—Fracture height vs. stage diagram.

variation among stages reflects the combined effects of local stress heterogeneity, rock mechanical properties, and injection parameters. Longer fractures are generally observed in stages with higher fluid volumes and less resistive lithologies.

Per-Cluster Equivalent Fracture Half-Length (Multifracture Model). In the multifracture model, fracture wave attenuation was analyzed and decomposed into multiple oscillatory components, each associated with a fracture cluster. The equivalent half-length of each cluster, derived based on energy partitioning and signal decay, is shown in Fig. 9. The values range from 76 m to 136 m, with a mean of 112 m. This distribution provides a refined view of the stimulation effectiveness across individual clusters and confirms that wave interference and stress shadowing reduce effective fracture extension.

Fracture Height. Fracture height estimates are summarized in Fig. 10, ranging from 26 m (Stage 18) to 47 m (Stage 9), with an average of 36 m. Unlike fracture length, the variation in fracture height is relatively limited. This suggests that vertical growth is likely governed by geological constraints, such as bedding planes, lamination density, or stress barriers, rather than by pumping volume alone. No statistically significant correlation was observed between fracture height and injection volume. Although the data in Fig. 10 may appear random, the overall trend aligns with the expected behavior of fracture propagation in heterogeneous reservoirs. The deviations from the mean values are relatively small, indicating that the observed patterns are not merely random noise but reflect the inherent complexity of the reservoir.

Correlation between Fracture Dimensions and Injection Parameters. To explore how injection parameters influence fracture geometry, statistical correlations between total injection volume and both fracture half-length and fracture height were evaluated. The results are shown in Fig. 11.

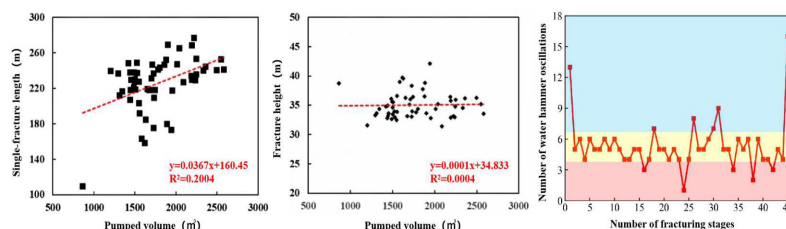


Fig. 11—(Left) Single-fracture length vs. pumped volume, (middle) fracture height vs. pumped volume, and (right) number of water hammer oscillations across 45 fracturing stages.

Single-Fracture Length vs. Pumped Volume. As shown in Fig. 11 (left), there is a moderate positive correlation between total injection volume and total fracture half-length. The Pearson correlation coefficient (r) is 0.47, and the Spearman rank correlation coefficient (ρ) is 0.48, indicating a consistent monotonic association. Pearson's correlation coefficient r measures the strength of linear correlation between two continuous variables, while Spearman's correlation coefficient ρ assesses the strength and direction of a monotonic (nonlinear or linear) relationship. In general, correlation coefficients between 0.4 and 0.6 are interpreted as moderate correlations (Mukaka 2012). However, the coefficient of determination (R^2) for linear regression is only 0.20, suggesting that injection volume alone explains a limited portion of fracture length variability. This result implies that while higher injection volumes may promote longer fractures, other factors—such as formation brittleness, stress anisotropy, and natural fracture systems—also significantly influence fracture propagation.

Fracture Height vs. Pumped Volume. In contrast, Fig. 11 (middle) shows that fracture height is not correlated with injection volume. The Pearson coefficient is 0.02, and the R^2 value is only 0.0004, indicating negligible linear dependence. This suggests that vertical fracture growth is not governed by pumping intensity but rather constrained by lithological layering, mechanical stratigraphy, and stress barriers that inhibit upward or downward propagation. The relative invariance in height across stages supports the hypothesis that vertical growth is geologically limited.

Number of Water Hammer Oscillations across 45 Fracturing Stages. To further investigate the controls on fracture propagation, we introduce Fig. 11 (right), which illustrates the number of water hammer oscillations detected across the 45 fracturing stages in A-Well. The average number of oscillations per stage is 5. Based on oscillation counts, the stages were classified into three categories: high complexity (≤ 4 oscillations, 15 stages), moderate complexity (5–6 oscillations, 25 stages), and low complexity (≥ 7 oscillations, 5 stages). Notably, regions with lower oscillation counts—indicating higher fracture complexity—tend to have shorter fracture lengths, even when the injection volumes are high. A correlation analysis between oscillation count and fracture half-length yields a Pearson's correlation

coefficient of -0.65 , indicating a moderate negative correlation. This supports the interpretation that fracture complexity, rather than injection volume alone, is a dominant control on fracture length in complex reservoir zones.

Correlation With Petrophysical Properties. To investigate the relationship between petrophysical properties and fracture propagation behavior, we performed Spearman's rank correlation analysis using logging-derived parameters and WHA-derived fracture metrics, including single-fracture length, per-cluster equivalent fracture half-length, fracture height, and the number of water hammer oscillations (used as a proxy for fracture complexity). The results are illustrated in Fig. 12.

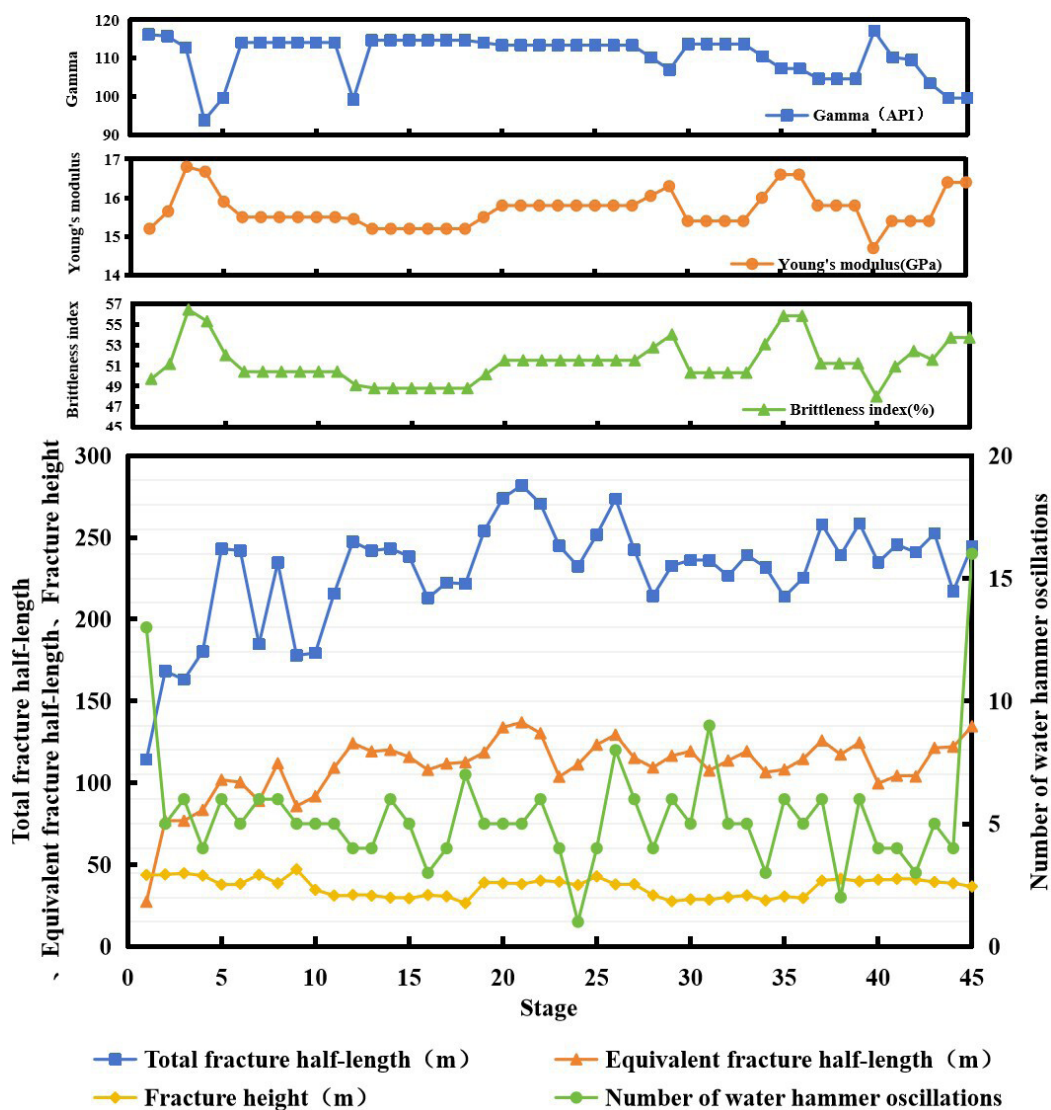


Fig. 12—Correlation between pressure oscillation count and petrophysical properties.

The gamma ray log (API), indicative of shale content, exhibits a moderate positive correlation with single-fracture length ($\rho = 0.561$), fracture height ($\rho = 0.486$), and oscillation count ($\rho = 0.401$), and a weaker positive correlation with per-cluster equivalent fracture half-length ($\rho = 0.287$). These relationships suggest that intervals with higher natural radioactivity may favor more extensive fracture growth and preserve more coherent pressure wave responses. A possible explanation is that bedding-parallel weaknesses in shale-rich zones promote vertical fracture containment and facilitate effective stimulation while also enhancing post-shut-in wave reflections due to improved fracture continuity.

In contrast, Young's modulus shows moderate negative correlations with single-fracture length ($\rho = -0.572$), per-cluster equivalent fracture half-length ($\rho = -0.462$), and fracture height ($\rho = -0.450$). This trend aligns with mechanical expectations that stiffer formations are more resistant to tensile opening and fracture extension. The weak negative correlation between Young's modulus and oscillation count ($\rho = -0.246$) further implies that higher-stiffness zones may reduce fracture connectivity and suppress pressure oscillation propagation.

The brittleness index, often regarded as a favorable indicator for fracture growth, shows weak to moderate negative correlations with fracture-related metrics: $\rho = -0.253$ for single-fracture length, $\rho = -0.145$ for per-cluster equivalent fracture half-length, and $\rho = -0.288$ for fracture height. Its correlation with oscillation count is minimal ($\rho = -0.093$). These results indicate that in this specific reservoir, higher brittleness does not consistently correspond to longer or more complex fractures. This may reflect the overriding influence of other geological factors such as stress anisotropy, natural fracture distribution, or proppant placement efficiency.

Finally, the number of water hammer oscillations, serving as a proxy for fracture complexity and post-shut-in wave energy retention, exhibits moderate positive correlations with single-fracture length ($\rho = 0.522$), per-cluster equivalent fracture half-length ($\rho = 0.435$), and fracture height ($\rho = 0.511$). This suggests that higher oscillation counts are more likely associated with larger and more continuous fracture

systems, rather than simply reflecting increased fracture tortuosity. In this case, pressure oscillations appear to represent fracture extent and coherence, not necessarily complexity in the classical sense.

Overall, these findings emphasize that fracture propagation and complexity in the studied formation are governed by interacting petrophysical and mechanical factors and that commonly used proxies such as brittleness may not always be reliable predictors on their own. The integrated use of fracture geometry data, petrophysical logs, and WHA-derived oscillation behavior offers a more pronounced understanding of stimulation effectiveness.

UTA Results. UTA was applied to evaluate the effective fracture surface area contributing to production during post-fracture flow. This method integrates both pressure and rate data to extract flow regime characteristics and diagnose the dynamic connectivity of the stimulated reservoir volume. Specifically, two time intervals were selected for analysis: Stage 1 (Days 48–51) representing the early linear flow regime and Stage 2 (Days 59–62) representing the later flow period. Slope fitting was applied to these segments to estimate and compare the effective fracture surface area A_f , as shown in Fig. 13. The variable τ enables the pressure response to be interpreted in a rate-normalized domain, facilitating flow regime identification and fracture surface estimation.

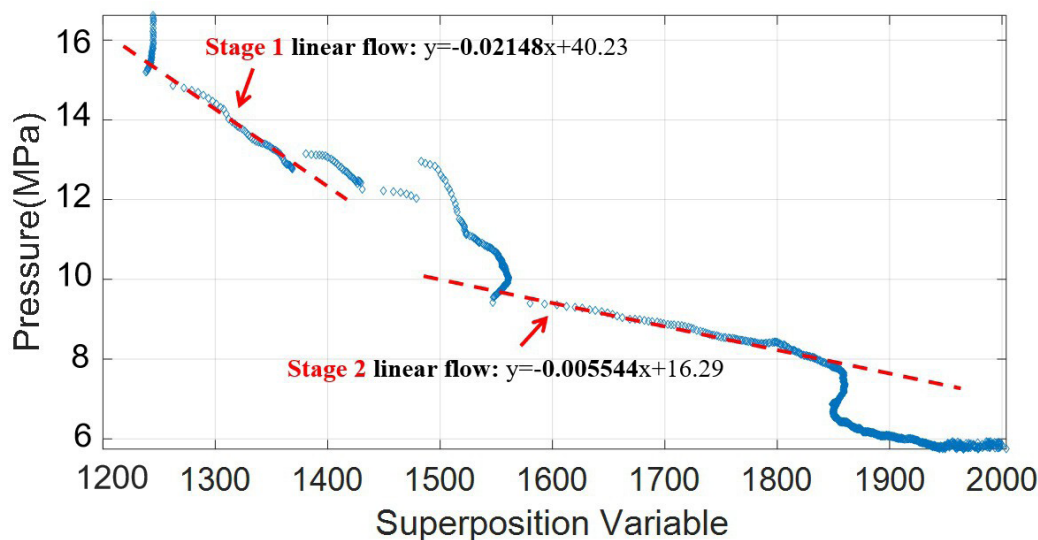


Fig. 13—Superposition plot of bottomhole pressure vs. superposition time variable for A-Well. Two analysis intervals are selected: Stage 1 (Days 48–51) and Stage 2 (Days 59–62), corresponding to early and later flow periods, respectively. These intervals are used for slope fitting to estimate effective fracture surface area A_f .

In Interval 1 (Days 48–51), the UTA-derived effective fracture area was calculated to be 231 538.91 m², while in Interval 2 (Days 59–62), the area increased to 752 668.45 m². This near-threefold increase illustrates the progressive activation of the fracture network over time. The difference reflects the evolution of reservoir pressure distribution and improved fracture connectivity as the reservoir system transitions from near-wellbore dominance to broader drainage. Early-time production is typically influenced by steeper pressure gradients and restricted flow near the wellbore, where fluid cleanup, capillary effects, or residual fracturing fluids may temporarily hinder conductivity. In contrast, later-time data represent a more stabilized flow regime with lower pressure gradients and enhanced communication with distal fracture segments. The observed increase in effective fracture area may be associated with improved drawdown efficiency with flowback of residual fracturing fluid and the mobilization of previously less-connected flow paths. While Fig. 13 alone does not directly capture pressure propagation depth, the evolution of A_f suggests an expansion of conductive fracture surface that could reflect enhanced fracture connectivity or delayed activation of distal branches (Deen et al. 2015).

Fig. 14 provides a comparison between fracture surface areas estimated by UTA and those derived from WHA. The total fracture surface area was calculated through WHA using $A_f = 2 \times \sum_{i=1}^{45} (L_i \times H_i)$, where L_i and H_i represent the half-fracture length (total or per-cluster sum) and height for each stage. The single-fracture model yielded a total area of 1 426 271.56 m², while the multifracture model yielded 1 404 359.69 m². These results represent the maximum geometrically stimulated region, regardless of long-term conductivity. The results also indicate that the single-fracture model and multifracture model yield similar results.

In comparison, the UTA-derived area from Interval 2 accounts for only 47% of the WHA-estimated total, revealing that a large portion of the hydraulically stimulated fracture network does not contribute meaningfully to production. This discrepancy underscores the importance of dynamic fracture evaluation: WHA captures the static mechanical footprint immediately after shut-in, while UTA reflects the evolving flow-effective volume that is both conductive and connected. Moreover, the similarity between fracture areas calculated using single- and multifracture WHA models implies that the half-length derived from the multifracture model per stage is approximately equal to the single-fracture model-derived half-length divided by the number of fracture clusters per stage. In other words, a single-fracture model can be used to estimate the average half-length per stage by using derived half-length divided by the total cluster number. Despite slight discrepancies due to modeling assumptions or input uncertainties, this result confirms the consistency of WHA geometry estimation under appropriate parameter selection.

Taken together, these findings highlight the diagnostic complementarity of UTA and WHA. UTA offers a time-dependent perspective of effective fracture contribution, critical for understanding stimulation efficiency. Operational strategies—such as choke tuning or flowback design—can be optimized using UTA insights to increase fracture utilization, reduce unproductive volume, and enhance overall recovery.

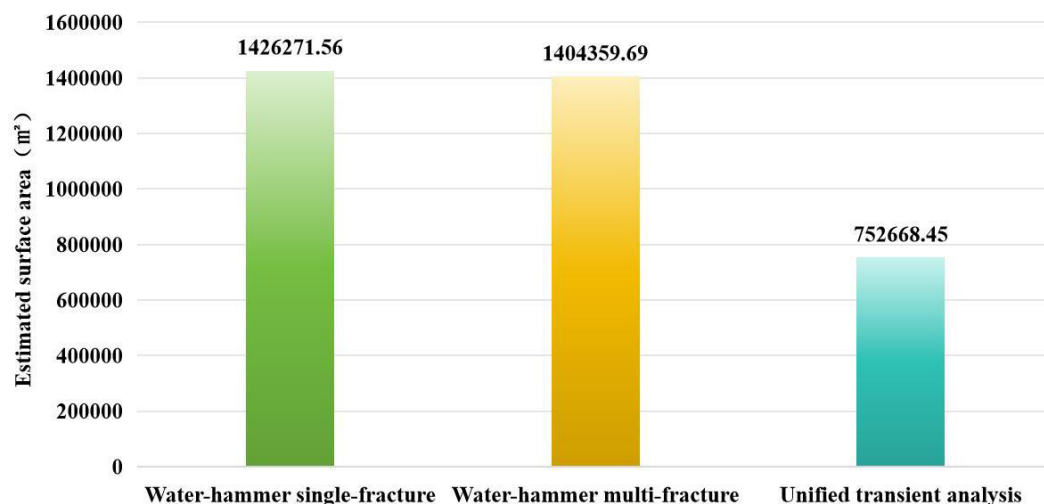


Fig. 14—Comparison of estimated surface between different methods.

Uncertainty Analysis. The discrepancy between fracture surface areas estimated by WHA and UTA can also arise from multiple sources of uncertainty, including model assumptions, measurement quality, reservoir heterogeneity, and fundamental differences in the physical behavior that each method captures. Understanding these limitations is critical to interpreting diagnostic results and guiding stimulation optimization.

Model Assumptions. WHA typically assumes planar, symmetrical, biwing fractures with uniform height, which may oversimplify the true geometry of multicluster fractures in horizontal wells. These assumptions can lead to overestimations of fracture dimensions, especially in zones affected by asymmetric growth or fracture interaction. UTA, by contrast, does not explicitly define fracture geometry but infers the effective contributing area from flow behavior, often assuming linear or bilinear flow regimes. When nonlinear flow, delayed connectivity, or progressive fracture closure is present, UTA may underestimate the area actively participating in production.

Measurement Errors. Both WHA and UTA are sensitive to data quality. Errors in bottomhole pressure reconstruction, pressure-dependent in fluid properties (e.g., viscosity and compressibility), and fluctuations in surface production rates or choke settings can introduce inaccuracies. In WHA, these uncertainties influence the estimation of fracture length and height from pressure oscillations. In UTA, they affect the identification of flow regimes and slope determination used to calculate effective fracture area.

Reservoir Heterogeneity. The Gulong shale formation exhibits substantial heterogeneity in lithology, rock properties, and in-situ stress distribution. These characteristics influence both fracture propagation and fluid flow paths, leading to spatial variability that cannot be fully captured by idealized models. As a result, the same diagnostic method may yield different results in different stratigraphic intervals, even under similar operational conditions.

Fundamental Physical Differences. WHA captures the total hydraulic footprint of the fracture system immediately after shut-in—specifically, all zones that responded to pressure wave propagation regardless of long-term conductivity. Because pressure waves travel rapidly and are unaffected by proppant distribution in the short term, WHA may reflect noncontributing or poorly connected fractures. In contrast, UTA reflects only the effectively contributing area—that is, portions of the fracture system that remain open, propped, and hydraulically connected over time. This distinction explains why UTA often produces smaller but more production-relevant estimates.

These differences are illustrated in this study, where the UTA-derived effective fracture area was approximately 47% of the area estimated by WHA (Fig. 14). This implies that less than one-half of the hydraulically stimulated volume ultimately contributes to production, providing a practical metric for evaluating proppant placement efficiency and fracture utilization. Such ratios can help diagnose treatment quality and guide decisions on refracturing or well spacing. These findings are consistent with previous studies. For example, Xu et al. (2015) used a material balance approach based on flowback data and found that the recovered volume was significantly less than the estimated created volume. Similarly, Mayerhofer et al. (2010) reported that although microseismic mapping could delineate the total stimulated reservoir volume, only a fraction of that volume was conductive and contributed to sustained flow, due to nonuniform proppant distribution, insufficient fracture conductivity, or disconnected flow paths. In summary, integrating WHA and UTA provides a more comprehensive understanding of fracture effectiveness. WHA defines the spatial extent of initial stimulation, while UTA evaluates its productive impact. These uncertainties are rooted in specific geological and engineering conditions. For example, strong stress contrasts between layers or the presence of natural fractures can lead to asymmetric fracture growth, which deviates from the planar assumptions of WHA. Proppant distribution in high-viscosity fluids may result in uneven conductivity, affecting UTA-derived area estimates. Surface flow rate fluctuations or choke control changes introduce signal noise that can interfere with pressure derivative stability. Additionally, the estimation of R , C , and I parameters is highly sensitive to the shut-in response duration and signal resolution. Recognizing these dependencies allows engineers to identify high-risk uncertainty sources and improve operational planning, such as using a supervisory control and data acquisition system for automatic flow measurement, tailoring injection, and shut-in schedules, and ensuring high-fidelity pressure data acquisition. This low-cost dual approach enhances diagnostic reliability and supports more effective HF design and reservoir management in unconventional formations.

Based on the combined insights from WHA and UTA, a practical workflow for stimulation evaluation and optimization is proposed, as illustrated in Fig. 5. The workflow begins with the acquisition of high-frequency shut-in pressure and production rate data. WHA is applied to interpret fracture complexity and estimate the total stimulated area immediately after shut-in. UTA is used to quantify the effective fracture surface area over time during production. These two diagnostic methods are applied in parallel and then integrated to identify discrepancies between stimulated and productive fracture surface area. The comparison of WHA-derived and UTA-derived areas provides a quantitative measure of fracture utilization. These results are then correlated with geological (such as logging data and seismic data) and design parameters (such as cluster spacing, perforation number, injection rate, fracturing fluid viscosity, and injection volume). Based on the diagnosis, treatment designs can be iteratively refined to improve proppant placement, fracture connectivity, and overall

stimulation efficiency (e.g., iteratively optimize design parameters for a certain geological formation to achieve maximum effective fracture surface area and/or to maximize the ratio of WHA-derived fracture surface area/UTA-derived fracture surface area).

Conclusions

With this study, we present a dual-method framework that integrates WHA and UTA to evaluate hydraulic-fracture effectiveness in the Gulong shale oil reservoir. WHA captures the initial extent and complexity of the fracture network immediately after shut-in while fractures area open, whereas UTA quantifies the long-term contributing surface area during production while fracture area is closed. The combination of these two diagnostics offers a more complete picture of both stimulation scale and productive impact.

Field application demonstrates that the UTA-derived effective fracture area was approximately 47% of the WHA-estimated area, highlighting a significant discrepancy between the total stimulated volume and the portion that ultimately contributes to production. This finding emphasizes the importance of evaluating not only the created fracture geometry but also its sustained conductivity.

The proposed integrated approach is cost-effective, nonintrusive, and relies solely on routinely acquired pressure and production data. A practical workflow is established for stimulation optimization, including data acquisition, diagnostic interpretation, and treatment design feedback. By applying this framework across multiple wells, operators can optimize HF design iteratively to maximize stimulation effectiveness and improve well performance.

Nomenclature

a	= pressure wave velocity, m/s
A	= cross-sectional or surface area, m ²
A_f	= effective fracture surface area (single wing), m ²
$A_{f,UTA}$	= fracture area estimated from unified transient analysis, m ²
$A_{f,WHA}$	= fracture area estimated from water-hammer analysis, m ²
B	= formation volume factor or pipe impedance, dimensionless
C	= capacitance in RCI model, m ³ /Pa
C^+	= positive characteristic line
C^-	= negative characteristic line
C_t	= total compressibility, 1/MPa
d_{ij}	= distance between fractures, m
D	= inner diameter of pipe, m
e	= pipe wall thickness, mm
E	= Young's modulus, GPa
E'	= plane strain elastic modulus, GPa
$E(m)$	= elliptic integral of second kind
f	= friction factor, dimensionless
g	= gravitational acceleration, m/s ²
h_f	= fracture height, m
H	= piezometric head, m
I	= inductance in RCI model, Pa·s ² /m ⁶
K	= formation permeability, m ²
L_f	= fracture half-length, m
n	= subsection index of pipe
P	= pressure, Pa or MPa
P_{bh}	= bottomhole pressure, Pa or MPa
P_f	= average fracture pressure or net pressure, Pa
ΔP_{mwf}	= near-wellbore friction loss, Pa
Q	= flow rate, m ³ /s
Q_0	= flow rate before shut-in, m ³ /s
R	= resistance in RCI model, Pa·s/m ³
Re	= Reynolds' number
S	= skin factor
S_{Hmin}	= minimum horizontal stress, MPa
t	= time, seconds or days
ν	= Poisson's ratio
V	= fluid velocity, m/s
W_f	= fracture width from viscous dissipation, mm
x	= spatial coordinate, m
z	= elevation above datum, m
λ	= multiplier or regression coefficient, dimensionless
μ	= fluid viscosity, mPa·s
ρ	= fluid density, kg/m ³
τ	= superposition time variable, seconds
ψ	= formation porosity, dimensionless

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